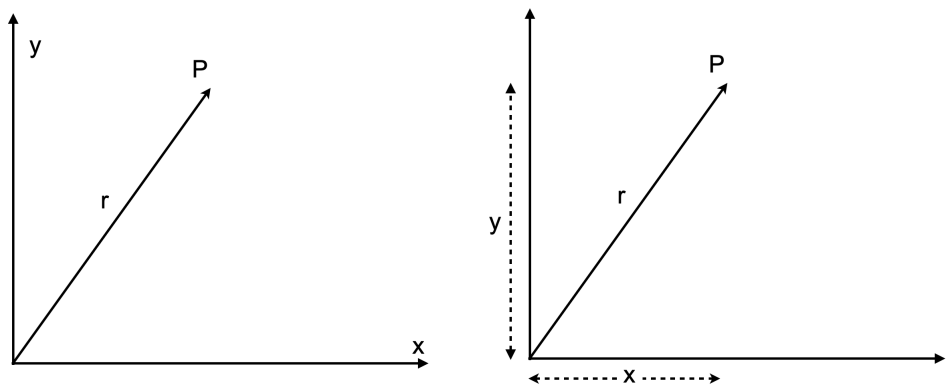
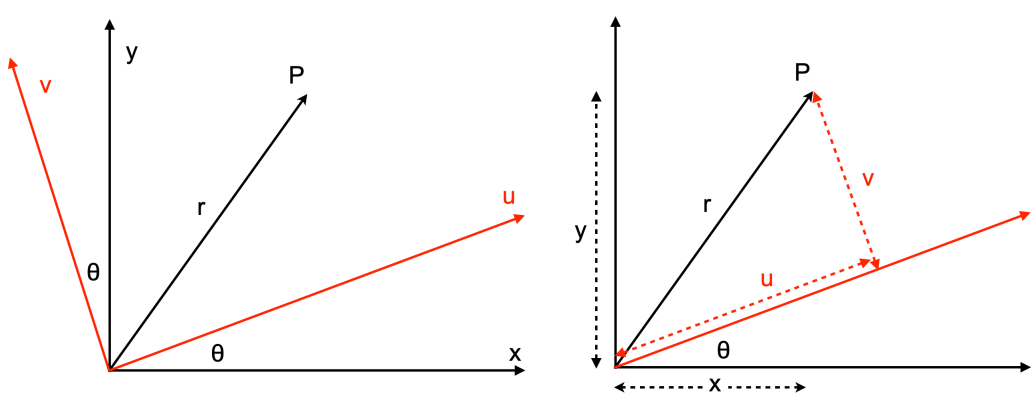
Rotation

Start with the standard picture from analytical geometry, courtesy of Fermat and Descartes. Pick a point for the origin. Draw orthogonal x- and y-axes.

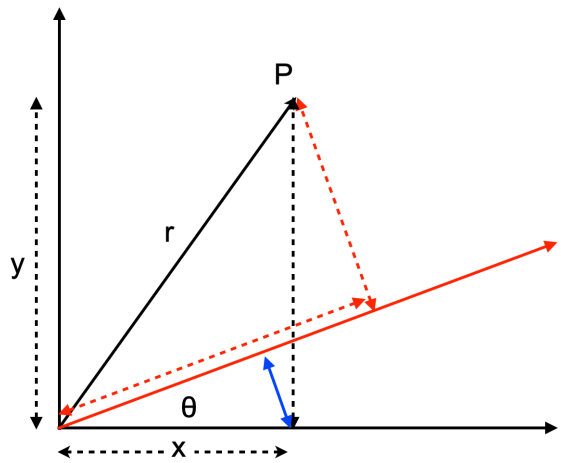


The point P lies a distance r from the origin. In the xy-coordinate system, $P = (x, y)$ and $r^2 = x^2 + y^2$.

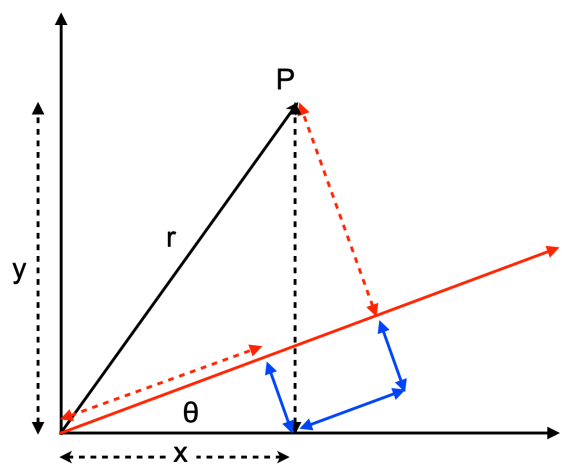


Now consider a second coordinate system with u- and v-axes, rotated counter-clockwise by an angle θ. In the new system, $P = (u, v)$.

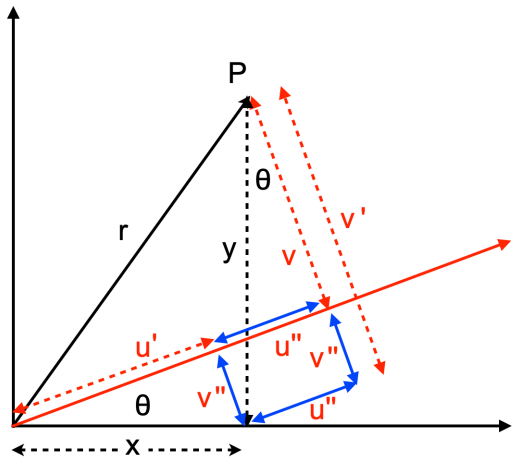
We will find a formula to give u as a function of x and y, that is $u = f(x, y)$. Simlarly, $v = g(x, y)$. (We already know that the length r must be the same in both systems so $x^2 + y^2 = u^2 + v^2$).



The first and key step is to draw a line that makes x the *hypotenuse* of a right triangle. That's the blue line in the figure above. Now, just fill in a rectangle with sides parallel to the u, v-axes.



Notice θ is the angle between y and v axes (top). Label segments of lines.



$$u' = x\cos\theta \qquad u'' = y\sin\theta$$
$$v' = y\cos\theta \qquad v'' = x\sin\theta$$

We can therefore write u as the sum of two segments:

$$u = u' + u'' = x\cos\theta + y\sin\theta$$

We write v as the difference of two segments.

$$v = v' - v'' = y\cos\theta - x\sin\theta$$

A convenient notation is to put them in a matrix that multiplies a vector:

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

This is R_{ccw}, the matrix for counter-clockwise rotation of coordinates, (from x, y to u, v).

Another view is that this is a clockwise rotation of the point P.

How about rotation in the other direction? This amounts to finding functions that turn u, v into x, y. It's easy when you know how. We have:

$$x = u\cos t - v\sin t$$

$$y = u\sin t + v\cos t$$

We manipulate the equations so that the second term on the right-hand side is identical but with opposite sign. Then just add.

Multiply the first equation by $\cos t$ and the second by $\sin t$. Addition gives:

$$x\cos t + y\sin t = u(\cos^2 t + \sin^2 t) - v\sin t\cos t + v\sin t\cos t$$

$$= u$$

Multiply the first equation by $-\sin t$ and the second by $\cos t$ and add

$$-x\sin t + y\cos t = v$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

The matrix above is R_{cw}.

If you know about matrices, you will realize that the two matrices must be inverses, since rotating first one direction and then the other leaves the point unchanged.

$$R_{ccw}\ R_{cw} = I$$

To invert a 2×2 matrix, transpose it and divide by the determinant (which is 1 here). So

$$
\begin{aligned}
R_{ccw} &= \frac{1}{|R_{cw}|} \cdot R_{cw}^T = R_{cw}^T \\
&= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}
\end{aligned}
$$

We can also check to make sure that these work correctly with the unit vectors. Recall that rotation of a point CCW is the same as rotation of the coordinate system CW, so we use R_{cw}:

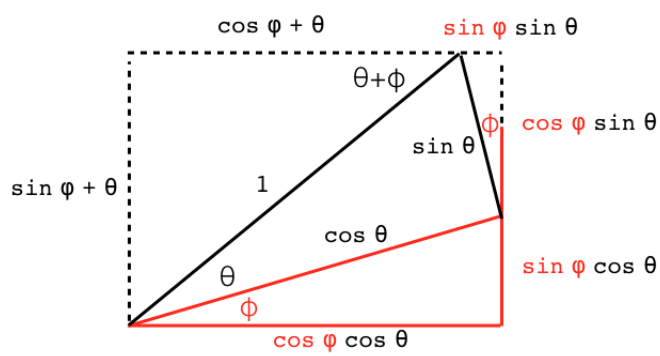
$$
\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}
$$

$$
\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}
$$

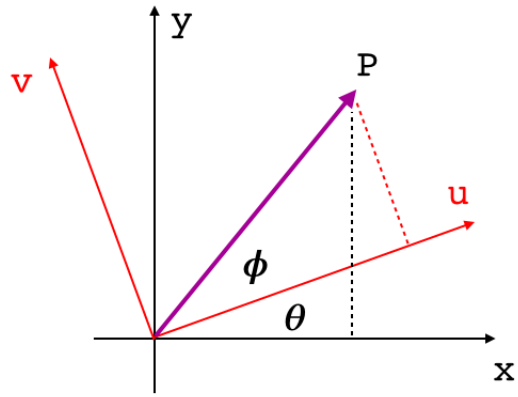
The rotated unit vector $\hat{\mathbf{j}}$ lies in the second quadrant.

Stewart

Probably the easiest method I have I found, in Stewart, depends on knowing the sum of angles formulas.



It goes like this.



Let the ray to the point P have length r. The coordinates of point P in the u, v system are naturally expressed in terms of ϕ:

$$u = r\cos\phi$$

$$v = r\sin\phi$$

while x and y are naturally expressed in terms of the combined angle $\theta + \phi$.

$$x = r\cos(\theta + \phi)$$

$$y = r\sin(\theta + \phi)$$

Now, use the sum formula for cosine:

$$x = r\cos\theta\cos\phi - r\sin\theta\sin\phi$$

Plug in from above:

$$x = u\cos\theta - v\sin\theta$$

It's as easy as that.

In the same way:

$$y = r\sin(\theta + \phi)$$

$$= r\sin\theta\cos\phi + r\cos\theta\sin\phi$$
$$= u\sin\theta + v\cos\theta$$

Notice that here we have x, y in terms of u, v. So this derivation naturally gives R_{cw}.

another view of CCW and CW rotation

Given $x = f(x, y)$ and $y = g(x, y)$, we wish to convert these to $u = f'(x, y)$ and $v = g'(x, y)$

One way is to switch symbols x, y for u, v, and at the same time, substitute $-\theta$ for θ. What this amounts to is relabeling our diagram with x, y being the rotated axes, and then rotating in the opposite direction (cw rather than ccw).

$$x = u\cos\theta - v\sin\theta$$

switch

$$u = x\cos -\theta - y\sin -\theta$$
$$= x\cos\theta + y\sin\theta$$

(Recall that $\cos -x = \cos x$ and $\sin -x = -\sin x$).

For y

$$y = u\sin\theta + v\cos\theta$$

switch

$$v = x\sin -\theta + y\cos -\theta$$
$$v = -x\sin\theta + y\cos\theta$$

vectors

Finally, another way is to guess. Start from vectors and matrices and remember that we have a 2×2 matrix.

Suppose the angle is $\theta = 0$. Then we want the matrix to be the identity matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

That naturally means that $\cos\theta$ is for entries along the diagonals and $\sin\theta$ at the other positions.

We also want to rotate the unit vector $\hat{\mathbf{i}}$, to become unit vector $\hat{\mathbf{j}}$.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

That looks good. So

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

means that

$$u = x\cos\theta - y\sin\theta$$

$$v = x\sin\theta + y\cos\theta$$

Rotating the points counter-clockwise is equivalent to rotating the coordinate system clockwise.

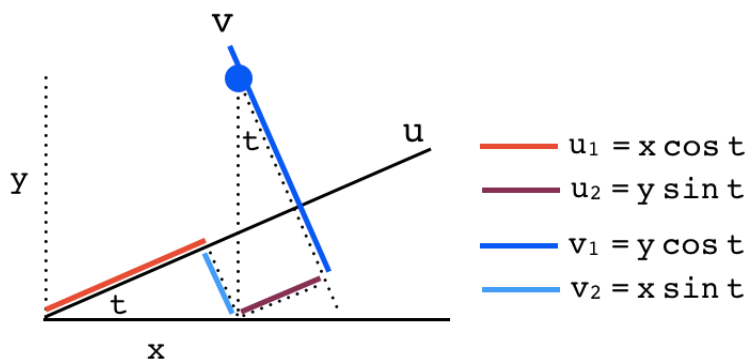
rotation by $+\theta$ then $-\theta$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

What follows is my original version of the first derivation.

standard derivation

Here is a second derivation with a sort of minimalist diagram of rotation



We draw the horizontal x-axis and the rotated u-axis. The angle between them is t. We plot our point and then draw perpendiculars to both axes. To finish the set-up, we draw perpendiculars from the point $(x, 0)$ as shown.

Once the drawing is rendered, we are almost done. Now we just work our way through.

$$u = u_1 + u_2 = x\cos t + y\sin t$$

All the triangles in the diagram are similar, with small angle t.

$$v = -v_2 + v_1 = -x\sin t + y\cos t$$

I find this hard to remember. You will know that you've done it right if you have both x and y as the *hypotenuse of a right triangle*. Start with the one for x and then rotate the triangle the other direction about the point $(x, 0)$. Stretch to fit.

Here we are rotating the coordinate system counter-clockwise. Hence, the minus sign is in the formula for the *vertical* component, v.

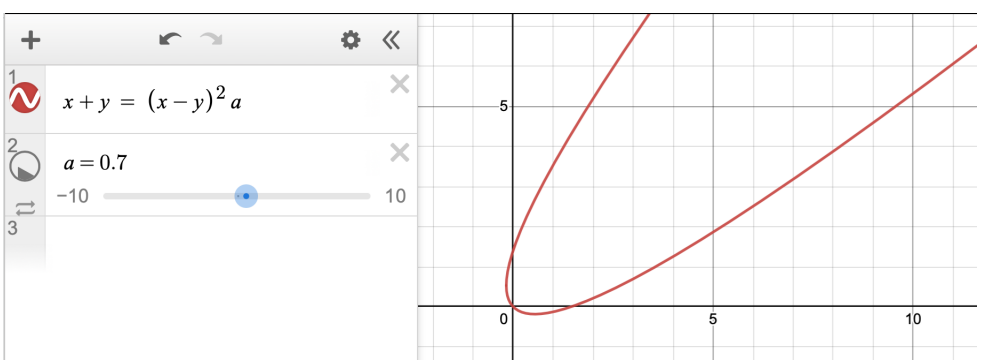
examples

Consider $y = x^2$ and rotate through an angle of 45° cw. The points are rotated cw and the coordinate system ccw, so the minus sign is in the expression for x'.

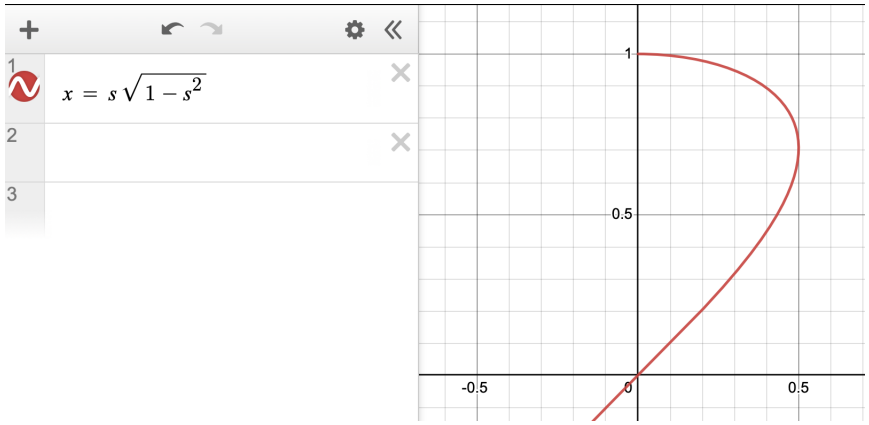
$$x' = 1/\sqrt{2}(x - y)$$
$$y' = 1/\sqrt{2}(x + y)$$

so

$$y' = x'^2$$
$$1/\sqrt{2} \cdot (x + y) = 1/2 \cdot (x - y)^2$$
$$x + y = 1/\sqrt{2} \cdot (x - y)^2$$



Here's another form — both rotated and translated.



Going back to our first result:

$$x' = x\cos\theta - y\sin\theta$$

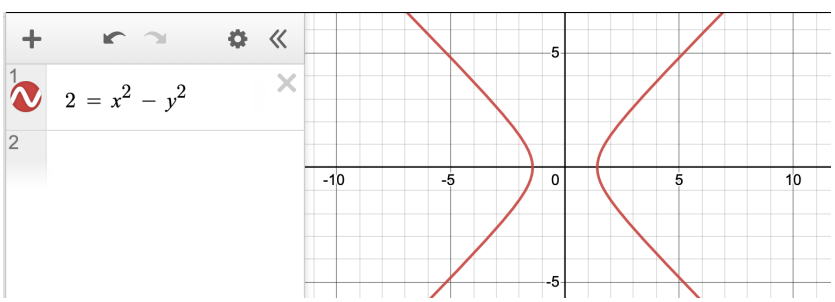
$$y' = x\sin\theta + y\cos\theta$$

Let $x'y' = 1$, that is, the graph is a hyperbola. We turn through an angle of 45° cw.

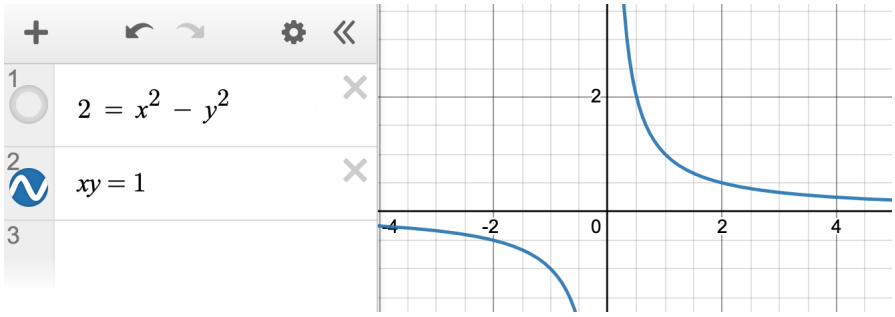
$$(x\cos\theta - y\sin\theta)(x\sin\theta + y\cos\theta)$$

$$\frac{1}{2}\cdot(x-y)(x+y)$$

$$2 = (x^2 - y^2)$$



When this guy is rotated by 45 ccw, it becomes the familiar graph of $xy = 1$.


1
$2 = x^2 - y^2$
2
$xy = 1$
3
2
-4
-2
0
2
4